

Ibn Tofail University

Algebra IV

Problem Set III

Exercise 1:

Let $A = \begin{pmatrix} -2 & 2 & -1 \\ -1 & 1 & -1 \\ -1 & 2 & -2 \end{pmatrix} \in M_3(\mathbb{R})$ and $B = \begin{pmatrix} -1 & 1 & 1 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$.

1. Determine the characteristic polynomial of A , then a basis of each sub-space.
2. Give the Jordan reduction J of A , the minimal polynomial of A , and the invertible matrix P such that : $A = PJP^{-1}$
3. Calculate J^n and deduce A^n .
4. Prove that A and B are the same.

Correction

Exercise 2:

Let

$$A = \begin{pmatrix} -4 & 1 & 0 & 1 \\ -2 & -1 & 0 & 1 \\ -12 & 6 & 3 & 1 \\ -2 & 1 & 0 & -1 \end{pmatrix}$$

1. Determine the eigenvalues of A .
2. Determine the dimensions of the sub space associated to the eigenvalues of A .
3. find the Jordan reduction matrix of A .
4. what is the minimal polynomial of A .

Correction

Exercise 3:

Let $A \in M_8(\mathbb{C})$ and it's minimal polynomial is $X^2(X-1)^3$ and the characteristic polynomial $X^4(X-1)^4$. Let J be the Jordan reduction of A .

What's the possible formes of J ?

Correction**Exercise 4:**

Let

$$A = \begin{pmatrix} 1 & \alpha & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 2 & 3 & 1 \\ -2 & -(4+\alpha) & -1 & \end{pmatrix}$$

where $\alpha \in \mathbb{R}$.

Give J and P Based on α . ($A = PJP^{-1}$).

Correction**Exercise 5:**

1. Solve the defferential system of 1st order :

$$\begin{cases} x'_1(t) = x_1(t) + x_2(t) \\ x'_2(t) = x_2(t) + x_4(t) \\ x'_3(t) = x_2(t) + x_3(t) \\ x'_4(t) = -x_1(t) + x_3(t) + 2x_4(t) \end{cases}$$

2. Solve the defferential equation of the 3th order :

$$x_t^3 = -x_t^2 + x_t^1 + x_t + t$$

Correction**Exercise 6:**

Let

$$A = \begin{pmatrix} 2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 \end{pmatrix}$$

1. Give the minimal polynomial of A .
2. Give the schema of the basis of Jordan of A , deduce the dimension of $\text{Ker}(A - 2I_7)^r$ where $r \in \mathbb{N}^*$.

Correction